```
!pip install kaggle
```

```
Requirement already satisfied: kaggle in /usr/local/lib/python3.12/dist-packages (1.7.4.5)
Requirement already satisfied: bleach in /usr/local/lib/python3.12/dist-packages (from kaggle) (6.2.0)
Requirement already satisfied: certifi>=14.05.14 in /usr/local/lib/python3.12/dist-packages (from kaggle) (2025.8.3)
Requirement already satisfied: charset-normalizer in /usr/local/lib/python3.12/dist-packages (from kaggle) (3.4.3)
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (from kaggle) (3.10)
Requirement already satisfied: protobuf in /usr/local/lib/python3.12/dist-packages (from kaggle) (5.29.5)
Requirement already satisfied: python-dateutil>=2.5.3 in /usr/local/lib/python3.12/dist-packages (from kaggle) (2.9.0.post0)
Requirement already satisfied: python-slugify in /usr/local/lib/python3.12/dist-packages (from kaggle) (8.0.4)
Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-packages (from kaggle) (2.32.4)
Requirement already satisfied: setuptools>=21.0.0 in /usr/local/lib/python3.12/dist-packages (from kaggle) (75.2.0)
Requirement already satisfied: six>=1.10 in /usr/local/lib/python3.12/dist-packages (from kaggle) (1.17.0)
Requirement already satisfied: text-unidecode in /usr/local/lib/python3.12/dist-packages (from kaggle) (1.3)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from kaggle) (4.67.1)
Requirement already satisfied: urllib3>=1.15.1 in /usr/local/lib/python3.12/dist-packages (from kaggle) (2.5.0)
Requirement already satisfied: webencodings in /usr/local/lib/python3.12/dist-packages (from kaggle) (0.5.1)
```

```
# configuring the path of Kaggle.json file
!mkdir -p ~/.kaggle
!cp kaggle.json ~/.kaggle/
!chmod 600 ~/.kaggle/kaggle.json
```

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# configuring the path of Kaggle.json file
!mkdir -p ~/.kaggle
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!chmod 600 ~/.kaggle/kaggle.json
```

```
# API to fetch the dataset from Kaggle
!kaggle datasets download -d omkargurav/face-mask-dataset
```

```
Dataset URL: https://www.kaggle.com/datasets/omkargurav/face-mask-dataset
License(s): unknown
Downloading face-mask-dataset.zip to /content
 91% 148M/163M [00:00<00:00, 296MB/s]
100% 163M/163M [00:00<00:00, 348MB/s]
```

```
# extracting the compessed Dataset
from zipfile import ZipFile
dataset = '/content/face-mask-dataset.zip'

with ZipFile(dataset,'r') as zip:
  zip.extractall()
  print('The dataset is extracted')
```

```
The dataset is extracted
```

```
ls
```

```
data/  face-mask-dataset.zip  kaggle.json  sample_data/
```

```
import os
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.image as mpimg
import cv2
from google.colab.patches import cv2_imshow
from PIL import Image
from sklearn.model_selection import train_test_split
import tensorflow as tf
from tensorflow import keras
```

```
with_mask_files = os.listdir('/content/data/with_mask')
print(with_mask_files[0:5])
print(with_mask_files[-5:])
```

```
['with_mask_1876.jpg', 'with_mask_950.jpg', 'with_mask_1334.jpg', 'with_mask_864.jpg', 'with_mask_2397.jpg']
['with_mask_1636.jpg', 'with_mask_1989.jpg', 'with_mask_2622.jpg', 'with_mask_2087.jpg', 'with_mask_2020.jpg']
```

```
without_mask_files = os.listdir('/content/data/without_mask')
print(without_mask_files[0:5])
print(without_mask_files[-5:])
```

```
['without_mask_1480.jpg', 'without_mask_691.jpg', 'without_mask_2453.jpg', 'without_mask_1944.jpg', 'without_mask_2582.jpg']
['without_mask_1273.jpg', 'without_mask_3251.jpg', 'without_mask_3366.jpg', 'without_mask_1758.jpg', 'without_mask_3298.jpg']
```

```
print('Number of with mask images:', len(with_mask_files))
print('Number of without mask images:', len(without_mask_files))
```

```
Number of with mask images: 3725
Number of without mask images: 3828
```

```
# create the labels

with_mask_labels = [1]*3725

without_mask_labels = [0]*3828
```

```
print(with_mask_labels[0:5])

print(without_mask_labels[0:5])
```

```
[1, 1, 1, 1, 1]
[0, 0, 0, 0, 0]
```

```
print(len(with_mask_labels))
print(len(without_mask_labels))
```

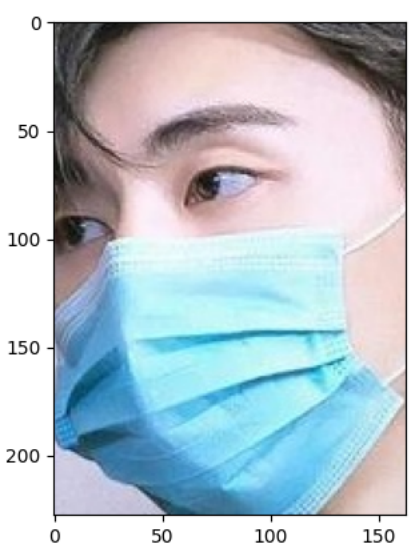
```
3725
3828
```

```
labels = with_mask_labels + without_mask_labels

print(len(labels))
print(labels[0:5])
print(labels[-5:])
```

```
7553
[1, 1, 1, 1, 1]
[0, 0, 0, 0, 0]
```

```
# displaying with mask image
img = mpimg.imread('/content/data/with_mask/with_mask_1545.jpg')
imgplot = plt.imshow(img)
plt.show()
```




```
# displaying without mask image
img = mpimg.imread('/content/data/without_mask/without_mask_2925.jpg')
imgplot = plt.imshow(img)
plt.show()
```

```
# convert images to numpy arrays+

with_mask_path = '/content/data/with_mask/'

data = []

for img_file in with_mask_files:

  image = Image.open(with_mask_path + img_file)
  image = image.resize((128,128))
  image = image.convert('RGB')
  image = np.array(image)
  data.append(image)



without_mask_path = '/content/data/without_mask/'


for img_file in without_mask_files:

  image = Image.open(without_mask_path + img_file)
  image = image.resize((128,128))
  image = image.convert('RGB')
  image = np.array(image)
  data.append(image)
```

```
/usr/local/lib/python3.12/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be cc
  warnings.warn(
```

```
# converting image list and label list to numpy arrays

X = np.array(data)
Y = np.array(labels)
```

```
type(X)
```

```
numpy.ndarray
```

```
type(Y)
```

```
numpy.ndarray
```

```
print(X.shape)
print(Y.shape)
```

```
(7553, 128, 128, 3)
(7553,)
```

```
X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size=0.2, random_state=2)

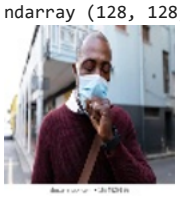

# scaling the data

X_train_scaled = X_train/255

X_test_scaled = X_test/255


X_train[0]
```

ndarray (128, 128, 3) show data



```
X_train_scaled[0]
```

```
array([[[1.        , 1.        , 1.        ],
        [1.        , 1.        , 1.        ],
        [1.        , 1.        , 1.        ],
        ...,
        [0.71372549, 0.78823529, 0.81568627],
        [0.71372549, 0.78039216, 0.80784314],
        [0.71372549, 0.78039216, 0.80784314]],

       [[1.        , 1.        , 1.        ],
        [1.        , 1.        , 1.        ],
        [1.        , 1.        , 1.        ],
        ...,
        [0.7254902 , 0.8       , 0.82745098],
        [0.71764706, 0.78431373, 0.81176471],
        [0.69411765, 0.76078431, 0.78823529]],

       [[1.        , 1.        , 1.        ],
        [1.        , 1.        , 1.        ],
        [1.        , 1.        , 1.        ],
        ...,
        [0.71372549, 0.78431373, 0.81568627],
        [0.69411765, 0.76078431, 0.79215686],
        [0.65098039, 0.71764706, 0.74509804]],

       ...,

       [[1.        , 1.        , 1.        ],
        [1.        , 1.        , 1.        ],
        [1.        , 1.        , 1.        ],
        ...,
        [1.        , 1.        , 1.        ],
        [1.        , 1.        , 1.        ],
        [1.        , 1.        , 1.        ]],

       [[1.        , 1.        , 1.        ],
        [1.        , 1.        , 1.        ],
        [1.        , 1.        , 1.        ],
        ...,
        [1.        , 1.        , 1.        ],
        [1.        , 1.        , 1.        ],
        [1.        , 1.        , 1.        ]],

       [[1.        , 1.        , 1.        ],
        [1.        , 1.        , 1.        ],
        [1.        , 1.        , 1.        ],
        ...,
        [1.        , 1.        , 1.        ],
        [1.        , 1.        , 1.        ],
        [1.        , 1.        , 1.        ]]])
```

```
num_of_classes = 2

model = tf.keras.Sequential()

model.add(tf.keras.layers.Conv2D(32, kernel_size=(3,3), activation='relu', input_shape=(128,128,3)))
model.add(tf.keras.layers.MaxPooling2D(pool_size=(2,2)))
```

```
model.add(tf.keras.layers.Conv2D(64, kernel_size=(3,3), activation='relu'))
model.add(tf.keras.layers.MaxPooling2D(pool_size=(2,2)))

model.add(tf.keras.layers.Flatten())

model.add(tf.keras.layers.Dense(128, activation='relu'))
model.add(tf.keras.layers.Dropout(0.5))

model.add(tf.keras.layers.Dense(64, activation='relu'))
model.add(tf.keras.layers.Dropout(0.5))


model.add(tf.keras.layers.Dense(num_of_classes, activation='sigmoid'))
```

```
/usr/local/lib/python3.12/dist-packages/keras/src/layers/convolutional/base_conv.py:113: UserWarning: Do not pass an `input_shape`/`inpu
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

```
# compile the neural network
model.compile(optimizer='adam',
              loss='sparse_categorical_crossentropy',
              metrics=['acc'])


# compile the neural network
model.compile(optimizer='adam',
              loss='sparse_categorical_crossentropy',
              metrics=['acc'])

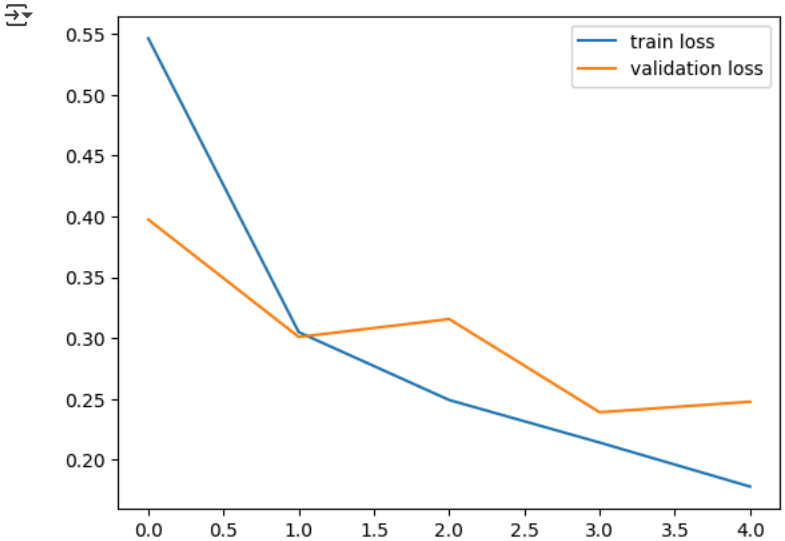
# training the neural network
history = model.fit(X_train_scaled, Y_train, validation_split=0.1, epochs=5)
```

```
Epoch 1/5
170/170 ━━━━━━━━━━━━━━━━━━━━ 157s 912ms/step - acc: 0.6526 - loss: 0.7843 - val_acc: 0.8364 - val_loss: 0.3975
Epoch 2/5
170/170 ━━━━━━━━━━━━━━━━━━━━ 198s 892ms/step - acc: 0.8785 - loss: 0.3266 - val_acc: 0.8810 - val_loss: 0.3010
Epoch 3/5
170/170 ━━━━━━━━━━━━━━━━━━━━ 203s 896ms/step - acc: 0.9045 - loss: 0.2471 - val_acc: 0.8777 - val_loss: 0.3157
Epoch 4/5
170/170 ━━━━━━━━━━━━━━━━━━━━ 209s 935ms/step - acc: 0.9151 - loss: 0.2260 - val_acc: 0.9256 - val_loss: 0.2391
Epoch 5/5
170/170 ━━━━━━━━━━━━━━━━━━━━ 152s 895ms/step - acc: 0.9321 - loss: 0.1842 - val_acc: 0.9124 - val_loss: 0.2478
```

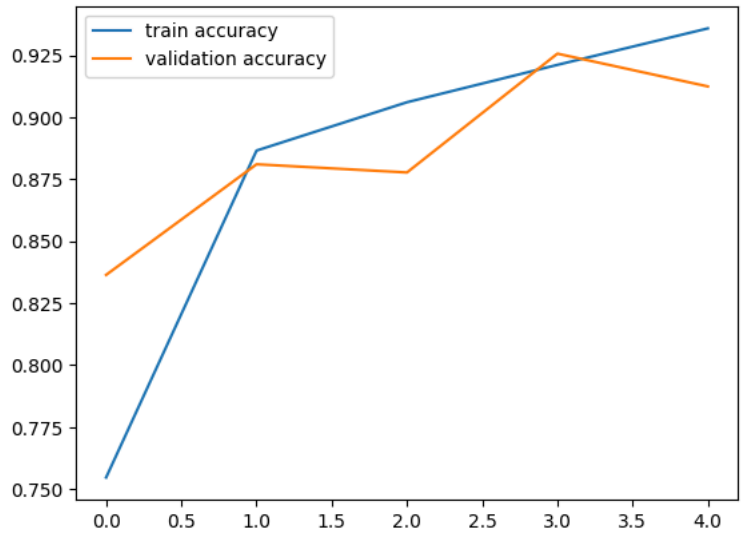
```
loss, accuracy = model.evaluate(X_test_scaled, Y_test)
print('Test Accuracy =', accuracy)
```

```
48/48 ━━━━━━━━━━━━━━━━━━━━ 28s 577ms/step - acc: 0.9164 - loss: 0.2373
Test Accuracy = 0.9179351329803467
```

```
# plot the loss value
plt.plot(history.history['loss'], label='train loss')
plt.plot(history.history['val_loss'], label='validation loss')
plt.legend()
plt.show()
```



```
# plot the accuracy value
plt.plot(history.history['acc'], label='train accuracy')
plt.plot(history.history['val_acc'], label='validation accuracy')
plt.legend()
plt.show()
```




```
input_image_path = input('Path of the image to be predicted: ')

input_image = cv2.imread(input_image_path)

cv2_imshow(input_image)

input_image_resized = cv2.resize(input_image, (128,128))

input_image_scaled = input_image_resized/255

input_image_reshaped = np.reshape(input_image_scaled, [1,128,128,3])

input_prediction = model.predict(input_image_reshaped)

print(input_prediction)


input_pred_label = np.argmax(input_prediction)

print(input_pred_label)


if input_pred_label == 0:

  print('The person in the image is wearing a mask')

else:

  print('The person in the image is not wearing a mask')
```

```
Path of the image to be predicted: /content/without mask.jpg
```

```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 55ms/step
[[0.1338136  0.72730994]]
1
The person in the image is not wearing a mask
```